

Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



Project Overview: Data & Goals

The Data Set

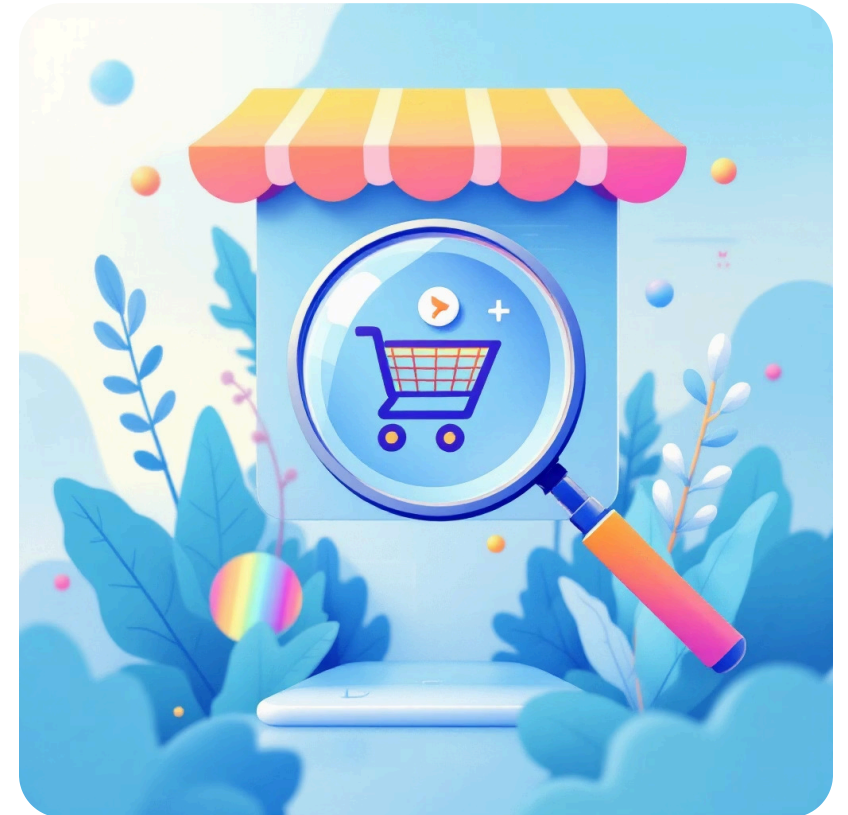
Transactional data from **3,900 purchases** across various product categories.

18 Columns

Demographics, Purchase Details, Shopping Behavior.

Key Insights

Spending patterns, customer segments, product preferences, and subscription behavior.



Exploratory Data Analysis (EDA) in Python

Initial steps focused on data preparation and cleaning for robust analysis.



Data Loading & Exploration

Imported dataset and checked structure/summary statistics.



Standardization & Engineering

Renamed columns to **snake case**. Created **age_group** and **purchase_frequency_days**.



Missing Data Handling

Imputed 37 missing Review Ratings using median by product category.



Database Integration

Loaded cleaned data into PostgreSQL for structured SQL analysis.

Key Financial Insights: Revenue & Spend

\$157K

Male Revenue

Total revenue generated by male customers.

\$75K

Female Revenue

Total revenue generated by female customers.

\$60.48

Express Shipping Avg. Spend

Higher average purchase amount than Standard Shipping (\$58.46).

SQL analysis revealed significant differences in revenue contribution and spending habits based on shipping preference.

Product Performance & Ratings

Top 5 Products by Average Rating

1. Gloves (3.86)
2. Sandals (3.84)
3. Boots (3.82)
4. Hat (3.80)
5. Skirt (3.78)

These products should be highlighted in marketing campaigns.

Discount-Dependent Products

Products with the highest percentage of discounted purchases (near 50% discount rate):

- Hat
- Sneakers
- Coat
- Sweater
- Pants



Subscription & Loyalty Analysis



Non-Subscribers Drive Revenue

2,847 non-subscribers generated **\$170K** in revenue (Avg. Spend: \$59.87).



Subscribers' Contribution

1,053 subscribers generated **\$62K** in revenue (Avg. Spend: \$59.49).

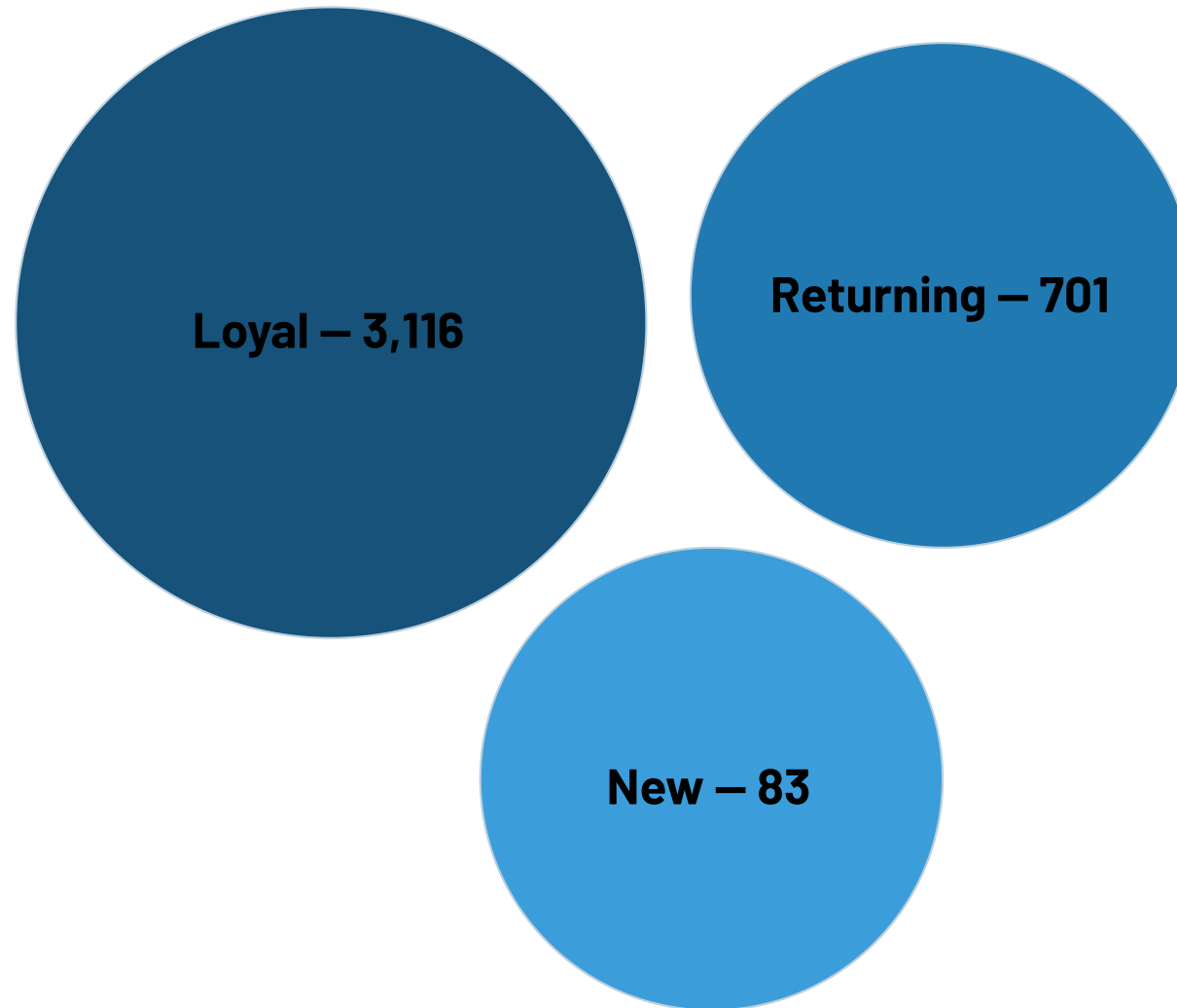


Repeat Buyers & Subscriptions

Customers with >5 purchases are **more likely** to subscribe (958 Yes vs. 2518 No).

Customer Segmentation

Classifying the customer base based on purchase history reveals a highly loyal core.



Loyal

3,116 Customers

Returning

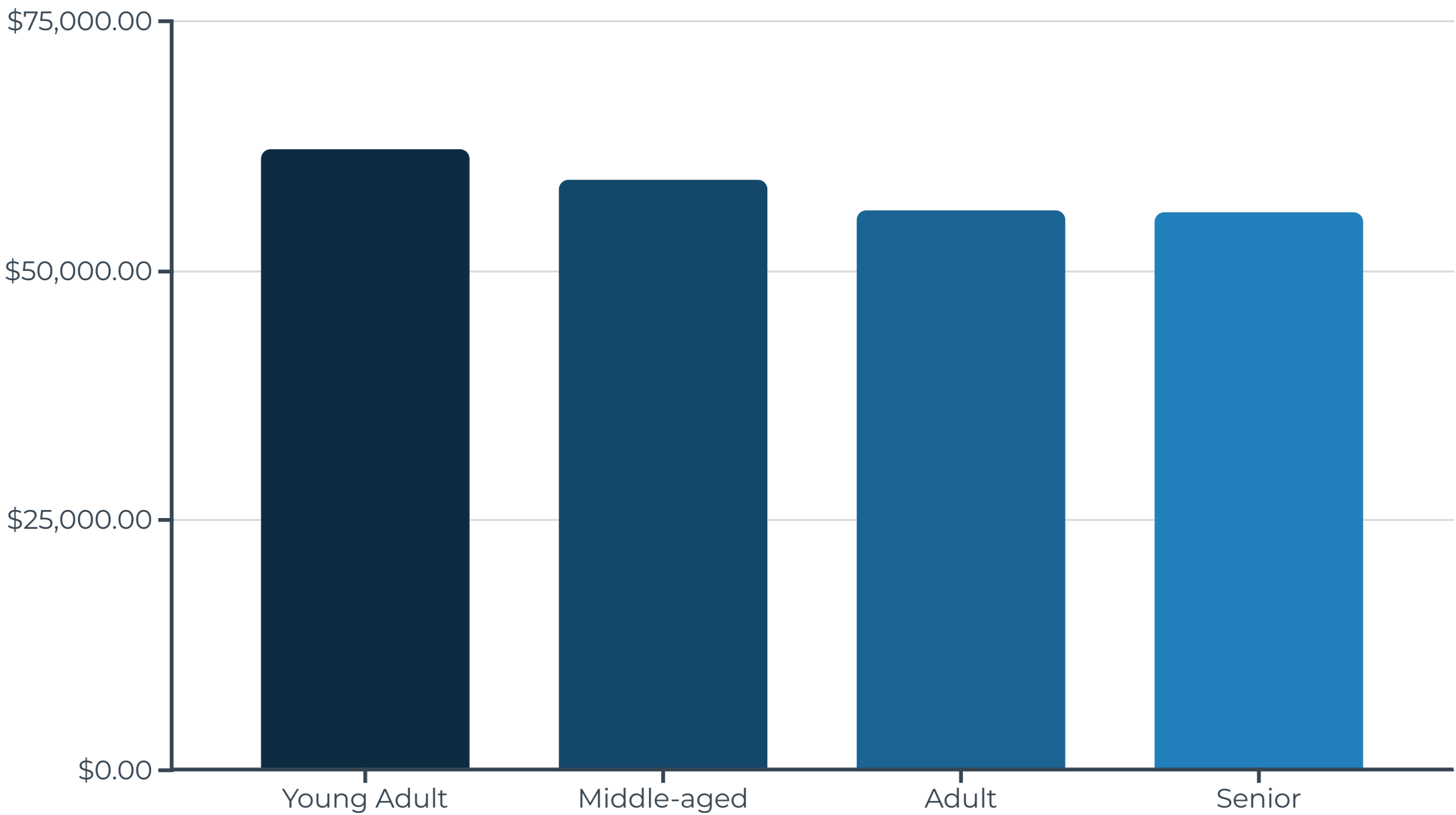
701 Customers

New

83 Customers

Revenue by Age Group

The highest revenue contribution comes from younger and middle-aged customers.



Targeted marketing should focus on the Young Adult and Middle-aged segments for maximum return.

Top Products Per Category

Identifying the best-selling items across the four main categories.



Accessories

Jewelry, Sunglasses, Belt



Clothing

Blouse, Pants, Shirt



Footwear

Sandals, Shoes, Sneakers



Outerwear

Jacket, Coat



Strategic Business Recommendations

→ **Boost Subscriptions**

Promote exclusive benefits to increase conversion among repeat buyers.

→ **Customer Loyalty Programs**

Reward repeat buyers to grow the highly valuable "Loyal" segment.

→ **Targeted Marketing**

Focus efforts on high-revenue age groups and Express Shipping users.

→ **Product Positioning**

Highlight top-rated and best-selling products in all campaigns.